Think — Project 2: clock rectangles

A Cambridge project: the numbers on a clock face, boxed four at a time, and the pattern that appears.

LESSON 71 | 1 × 40 MIN | MATEMATIKA 6, O‘YLAB KO‘R | S7 PROJECT 2

LESSON CLOCK

8' | 12' | 14' | 6'

The arrangement	8 min
Collecting the data	12 min
Stating the pattern	14 min
Explaining why	6 min

LEARNING OBJECTIVES

- Collect data systematically from a simple arrangement.
- Notice a pattern and state it in words.
- Express the pattern with letters.
- Test the statement on cases not yet tried.

TEXTBOOK

National	pp. 203–205
Cambridge	Stage 7, project pages
Workbook	Project sheet 2

01 Explanation

The arrangement

Write the numbers *1* to *12* in three rows of four, as on a calendar page:

1	*2*	*3*	*4*
5	*6*	*7*	*8*
9	*10*	*11*	*12*

Draw a rectangle round any four numbers that form a block — two rows and two columns — and multiply each pair of opposite corners.

THE QUESTION

What is the difference between the two products? Try several rectangles before deciding.

Collecting the data

RECTANGLE	ONE DIAGONAL	THE OTHER	DIFFERENCE
$1, 2, 5, 6$	$1 \cdot 6 = 6$	$2 \cdot 5 = 10$	4
$2, 3, 6, 7$	$2 \cdot 7 = 14$	$3 \cdot 6 = 18$	4
$6, 7, 10, 11$	$6 \cdot 11 = 66$	$7 \cdot 10 = 70$	4
$3, 4, 7, 8$	$3 \cdot 8 = 24$	$4 \cdot 7 = 28$	4

BE SYSTEMATIC, NOT RANDOM

Working across the top row and then down catches every case and makes it obvious when the pattern holds. Random examples leave gaps that hide exceptions.

Stating the pattern

The difference is always 4. Now say why, with letters: let the top-left number be n.

POSITION	NUMBER
top left	n
top right	$n + 1$
bottom left	$n + 4$
bottom right	$n + 5$

$$(n + 1)(n + 4) - n(n + 5) = n^2 + 5n + 4 - n^2 - 5n = 4$$

THE LETTERS PROVE WHAT THE NUMBERS SUGGESTED

Four examples show a pattern; the algebra shows it must hold for every rectangle in the grid, including ones nobody has tried.

Explaining why

GRID WIDTH	BOTTOM-LEFT IS	THE DIFFERENCE
4	$n + 4$	4
5	$n + 5$	5
7	$n + 7$	7
w	$n + w$	w

THE DIFFERENCE IS THE WIDTH OF THE GRID

A calendar month, seven columns wide, gives a difference of 7 for every 2×2 block — which is worth testing on a real calendar before the lesson ends.

02 Worked examples

EXAMPLE 1 **Take the rectangle $6, 7, 10, 11$. Find the two products and their difference.**

1. $6 \cdot 11 = 66$
2. $7 \cdot 10 = 70$
3. $70 - 66 = 4$

ANSWER

4

EXAMPLE 2 **Prove that the difference is always 4 in a grid four columns wide.**

1. Let the top-left number be n.
2. The four numbers are n, $n + 1$, $n + 4$, $n + 5$.
3. $(n + 1)(n + 4) = n^2 + 5n + 4$
4. $n(n + 5) = n^2 + 5n$, so the difference is 4.

ANSWER

Always 4

EXAMPLE 3 **What would the difference be on a calendar, seven columns wide?**

1. The bottom-left number is $n + 7$.
2. $(n + 1)(n + 7) = n^2 + 8n + 7$
3. $n(n + 8) = n^2 + 8n$
4. The difference is 7.

ANSWER

7

03 Interactive model

Hand out real calendar pages and let the class discover the seven before any algebra; the general result then answers a question they have already asked.

The clock-rectangle pattern

For $1, 2, 5, 6$ the difference is:

- 2
- 4
- 6
- 10

For $6, 7, 10, 11$ the difference is:

- 4
- 6
- 66
- 70

If the top left is n, the bottom right is:

- $n + 4$
- $n + 5$
- $n + 1$
- $n + 6$

$(n + 1)(n + 4)$ equals:

- $n^2 + 4$
- $n^2 + 5n + 4$
- $n^2 + 4n + 1$
- $n^2 + 5$

$n(n + 5)$ equals:

- $n^2 + 5$
- $n^2 + 5n$
- $n + 5n$
- $5n^2$

So the difference is:

- n
- 4
- 5
- $5n$

On a seven-column calendar the difference is:

- 4
- 7
- 12
- 49

Four examples showing the pattern are:

- a proof
- evidence for a conjecture
- irrelevant
- a counter-example

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Clock face	Soat siferblati	Циферблат
Rectangle	To‘rtburchak	Прямоугольник
Diagonal	Diagonal	Диагональ
Pattern	Qonuniyat	Закономерность
Conjecture	Faraz	Гипотеза
To generalise	Umumlashtirish	Обобщить
Systematic	Tartibli	Систематический
Proof	Isbot	Доказательство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

In the 3×4 grid the difference is always:

- 2
- 4
- 5
- it varies

With top left n, the bottom left is:

- $n + 1$
- $n + 4$
- $n + 5$
- $n + 3$

The proof needs:

- more examples
- letters
- a calculator
- a diagram only

On a grid w wide the difference is:

- 4
- w
- $2w$
- w^2

A pattern seen in four cases is:

- proved
- a conjecture
- false
- a theorem

Systematic data collection means:

- random examples
- working in order
- one example
- guessing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The difference for $1, 2, 5, 6$

   ANSWER 4

2. The difference for $2, 3, 6, 7$

   ANSWER 4

3. The difference for $3, 4, 7, 8$

   ANSWER 4

4. The difference for $6, 7, 10, 11$

   ANSWER 4

5. With top left n, the top right

   ANSWER $n + 1$

6. The bottom left

   ANSWER $n + 4$

7. The bottom right

   ANSWER $n + 5$

Medium · the standard question

7 PROBLEMS

1. $(n + 1)(n + 4)$ expanded

   ANSWER $n^2 + 5n + 4$

2. $n(n + 5)$ expanded

   ANSWER $n^2 + 5n$

3. Their difference

   ANSWER 4

4. On a five-column grid

   ANSWER 5

5. On a calendar (seven columns)

   ANSWER 7

6. On a grid w columns wide

   ANSWER w

7. Is the pattern proved by four examples?

   ANSWER No — they are evidence only

Hard · stretch and reason

7 PROBLEMS

1. A 3×3 block on a seven-column calendar: the corner difference

   ANSWER 28

2. Why does the n^2 term always cancel?

   ANSWER Both products have the same leading term

3. Predict the difference on a ten-column grid

   ANSWER 10

4. Test the prediction with $n = 3$ on a ten-column grid

   ANSWER $4 \cdot 13 - 3 \cdot 14 = 10$ ✓

5. What would happen with a 2×3 block on a four-column grid?

   ANSWER The difference becomes 8

6. Write the four numbers of a block with bottom right n

   ANSWER $n - 5, n - 4, n - 1, n$

7. Why is algebra better than more examples here?

   ANSWER It settles every case at once, including untried ones

07 Homework

Homework — the project

One page: your data table, the pattern in words, the algebra, and one prediction tested.

1. Draw the 3×4 grid and test six different rectangles.
2. Record your results in a table and state the pattern in one sentence.
3. Prove the pattern using n for the top-left number.
4. Repeat the investigation on a real calendar month and record what happens.
5. Predict the difference on a grid 9 columns wide, then test your prediction.